

Electron-Ion Collider Record of Decision

TITLE	Timing Basis for the EIC Machine Protection System Fast-Abort Requirement
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AFFECTED WBS/PROJECT AREA	WBS 6.04.04.03 IR – SIFI – Machine Protection and Collimation Systems, WBS 6.04.04.04 IR – SIFI – Controls Systems, WBS 6.02.02.05 ASR – ESR – Instrumentation Systems, WBS 6.02.03.05 ASR – HSR – Instrumentation Systems, WBS 6.04.03.06 IR – AS – Beam Instrumentation Systems
STATEMENT OF DECISION:	
The preliminary timing basis for the EIC MPS fast-abort function shall be an end-to-end response of no more than 4 turns for designated fast protection functions, measured from the first validated fast detection event to the point when the beam has reached a safe state.	

Description/Purpose:

This RoD establishes the preliminary timing basis to be used in the PRD for the Controls-facing MPS performance requirement. It records the system-level response time required from the **designated fast protection chain**. For a hazardous fast failure, the protection chain shall detect the event, evaluate the interlock condition, distribute the abort signal, actuate the abort system, and bring the beam to a safe state **within 4 turns**.

This timing requirement applies specifically to **fast failures** that can develop into equipment damage or unacceptable beam loss within a few turns. Typical examples include:

- a kicker misfire or kicker timing error that places beam on a protected aperture;
- a fast magnet PS trip or sudden current excursion that produces a rapid orbit or optics error;
- an RF trip, RF phase error, or cavity fault that generates fast off-momentum loss;
- a fast beam loss spike detected at a collimator or IR protection location;
- a rapid orbit excursion toward a tight aperture or sensitive element.

For such cases, the designated fast-abort path is defined as:

fast sensor or equipment permit → local interlock input → deterministic permit / abort logic → abort request / global permit drop → signal distribution to abort hardware → kicker / dump actuation → beam-safe condition

In practical terms, this means that a fast BLM may detect an abnormal loss burst, a BPM-based orbit interlock may detect a large excursion, or an RF or powering system may drop a critical permit. That signal is then processed in deterministic logic, the abort request is transmitted to the beam abort system, the kicker fires on the proper abort gap, and the circulating beam is removed before losses grow to a damaging level.

The **4-turn value** is based on the following preliminary budget and associated sub-subsystems for the designated fast-protection chain:

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Stage	Description	Latency (max)	Sub-system
Validated detection	Fast sensor + front-end response for verified fast inputs (e.g., fast BLM threshold crossing, fast orbit-limit output, fast RF/powering permit loss)	< 1 turn	Instrumentation Systems
Logic evaluation	Fixed-latency FPGA computation of the designated interlock/permit decision, including voting and machine-mode mapping	< 1 micro-s	Controls Systems
Transmission	Interlock signal propagation to ESR/HSR inhibit/abort actuation, including integrity checks	≈ 1 turn	
Actuation	Trigger handling (abort gap alignment) and kicker field rise (within the abort gap) to required value	≤ 1 turn	
Beam removal	Ring-dependent time to achieve beam-safe condition	= 1 turn	Machine Protection System
Total	End-to-end response time for designated fast protection functions	≤ 4 turns	

This gives a **total end-to-end response of no more than 4 turns**. Within this budget, the logic stage is very small compared with one revolution period, so the dominant contributors are the sensor response, transmission path, synchronization and actuation, and final beam removal.

The requirement recorded here is a **system-level timing basis** for the fast designated protection path. It does not imply that every machine input must respond on that timescale. Slow channels such as vacuum gauges, slow thermal readbacks, and other supervisory signals remain important for machine protection, but they are not designated for the **4-turn fast-abort requirement**. Likewise, some genuinely single-turn hazards may still require passive protection because active systems cannot react before the damage process begins. The purpose of this decision is to define the expected timescale for the active, deterministic, fast-abort path.

This turn-scale requirement is consistent with experience at comparable machines. At the LHC [<https://proceedings.jacow.org/ibic2022/papers/tup03.pdf>], beam protection literature reports that the **beam must be extracted in less than 3 turns** when excessive losses are detected, and that upgraded BLM electronics can **detect losses and request a beam dump in less than 1 turn**. At RHIC [<https://inspirehep.net/files/6dc00eae1de57941eb37a59f584f1081>, <https://proceedings.jacow.org/ipac2022/papers/thpost048.pdf>], delayed-abort studies showed that millisecond-scale latency is too long for a superconducting hadron ring: one study reports about 7 ms between removal of the beam permit and firing of the abort kickers, and a later upgrade paper reports operation with **7 ms abort-kicker delay, corresponding to 547 turns**. At SuperKEKB [<https://proceedings.jacow.org/ICALEPCS2015/papers/mopgf141.pdf>, <https://proceedings.jacow.org/ibic2016/papers/tual03.pdf>, <https://proceedings.jacow.org/ipac2023/pdf/MOPL072.pdf>], published abort-system and diagnostics studies support a few-turn to few-tens-of-microseconds protection timescale: the abort trigger upgrade reduced the response time from about 100 micro-s to less than **20 micro-s**, and more recent **sudden beam loss** studies describe beam loss developing **within a few turns**.

Taken together, these external examples support the engineering basis adopted here: microsecond-level logic alone is not sufficient; the full protection chain must be evaluated end-to-end, including detection, transmission, synchronization, kicker response, and actual beam removal. The preliminary

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decision for EIC is therefore to adopt ≤ 4 turns as the system-level performance basis for designated fast-abort functions in the PRD.

APPROVALS:

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Acronym List

RHIC	Relativistic Heavy Ion Collider
EIC	Electron-Ion Collider
ESR	Electron Storage Ring
HSR	Hadron Storage Ring
ASR	Accelerator Storage Ring
IR	Interaction Region
MPS	Machine Protection System
SIFI	System Installation and Final Integration
AS	Accelerator Systems
LHC	Large Hadron Collider
PS	Power supply
PDR	Preliminary Design Report
BLM	Beam Loss Monitor
BPM	Beam Position Monitor
FPGA	Field-Programmable Gate Array
RF	Radiofrequency
WBS	Work Breakdown Structure